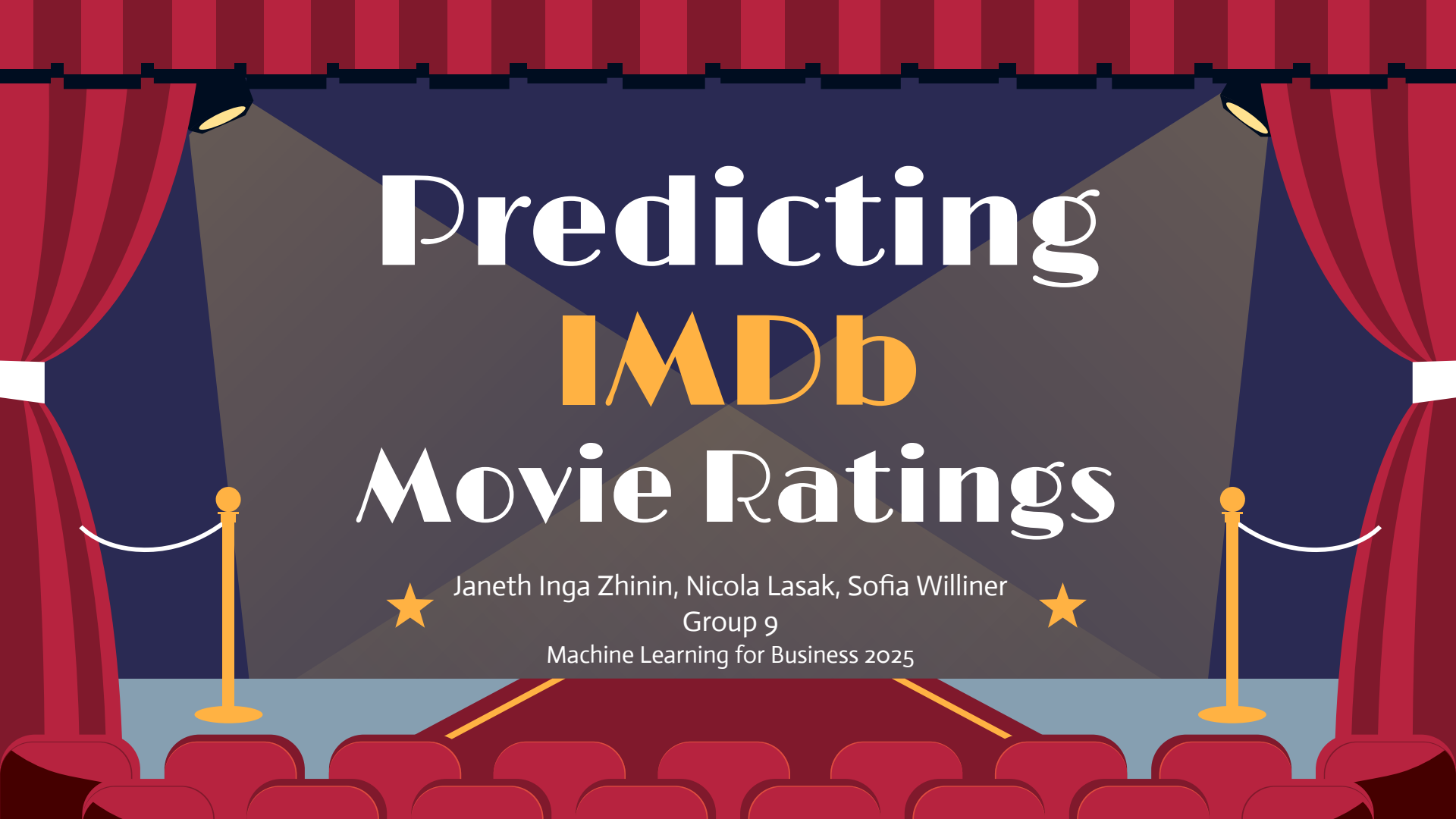




Predicting IMDb Movie Ratings

★ Janeth Inga Zhinin, Nicola Lasak, Sofia Williner ★

Group 9

Machine Learning for Business 2025

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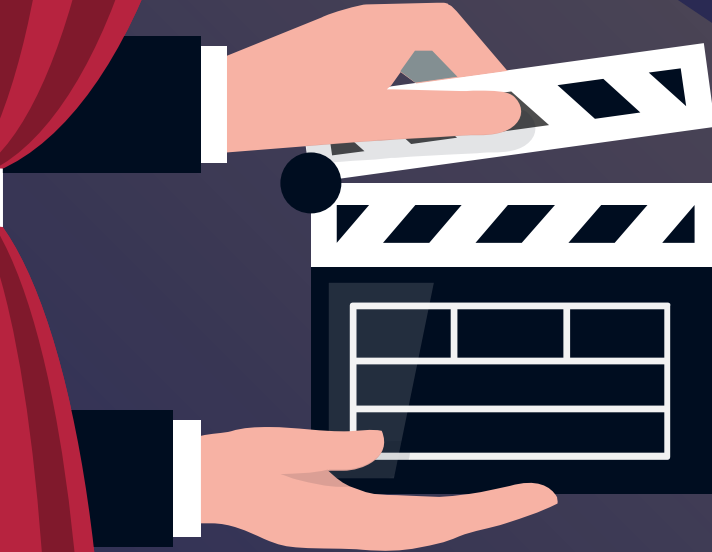
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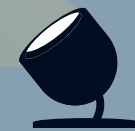
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An illustration of a hand holding a clapperboard, with a spotlight beam shining on it from the left. The clapperboard has a black top bar with white diagonal stripes, a black middle bar with white diagonal stripes, and a black bottom bar with white diagonal stripes. The hand is holding the clapperboard from the left side, with the thumb and index finger visible. The background is dark blue with red curtains on the left and right sides. A spotlight beam from the bottom right corner illuminates the clapperboard and the text area.

01

Introduction

Goal of the Project, Data Overview,
and Modeling Approach

A small black spotlight on a stand, positioned on the floor and pointing upwards towards the text area.

Goals



- ★ Construct a machine learning system that can predict a movie's IMDb rating based on creative, technical, and production characteristics
- ★ To identify which film attributes most strongly influence audience reception
- ★ To extract business insights about what drives a successful movie

Problem Definition:

Why predict movie ratings?

- ★ Movie ratings influence decisions for studios, streaming platforms, and marketers
- ★ Predicting ratings helps understand what makes a movie successful
- ★ Our project aims to analyze movie attributes (genre, runtime, budget, etc.) and predict audience ratings
 - Similar data can be used to predict movie success before official release dates
- ★ Helps bridge creative and data-driven decision-making



Data Processing Overview

- ★ TMDB 5000 Movies Dataset
 - Movies.csv and Credits.csv files
- ★ Preprocessed the data by parsing nested fields and extracting individual features to convert into structured features
 - Genres, crew, keywords, production companies, etc.
- ★ Used feature engineering to improve model accuracy and interpretability
 - Top genre, director, top keyword, top company, runtime, budget, revenue, vote count, etc.
 - Created feature table to use for modeling
- ★ Created targets per model
 - Regression → average vote
 - Classification → high-rated and low-rated threshold



MODELING APPROACH



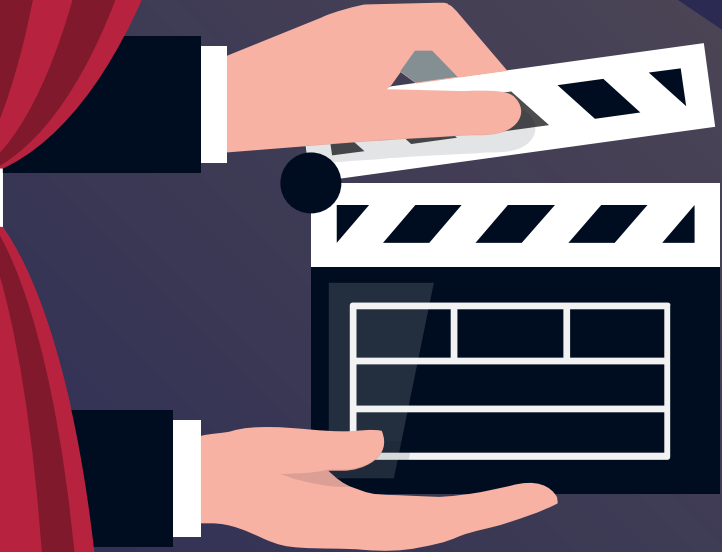
Classification

- ★ Predicts high-rated vs low-rated
- ★ Simple & interpretable
- ★ Features & outcomes = linear
- ★ Successful business use case



Ridge Regression

- ★ Predicts the exact IMDb rating
- ★ More detailed and interpretable predictions
- ★ Treats ratings as a continuous outcome
- ★ Supports business forecasting, budgeting, and performance

A stylized illustration of a hand holding a clapperboard. The hand is light pink with a black sleeve and white cuff. The clapperboard is white with black diagonal stripes on the top bar and a black rectangular area with white horizontal lines below. A black circular button is visible on the left side of the clapperboard. The background is dark blue with a spotlight effect from the top left, and red curtains are visible on the left and right sides.

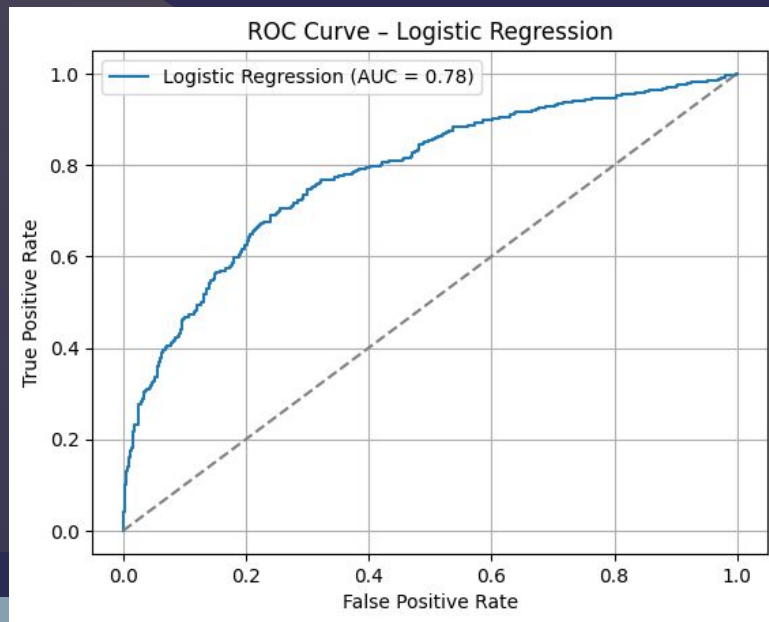
02

Phase 1

Classification Model,
Takeaways & Improvements

A small, black, stylized stage spotlight on a stand, located in the bottom right corner of the image.

Classification Model



Classification Model



Performance

- Balanced classes
- Strong ROC curve separation
- Clear positive predictors: Vote Count, Popularity, Certain genres/directors, Collection status.



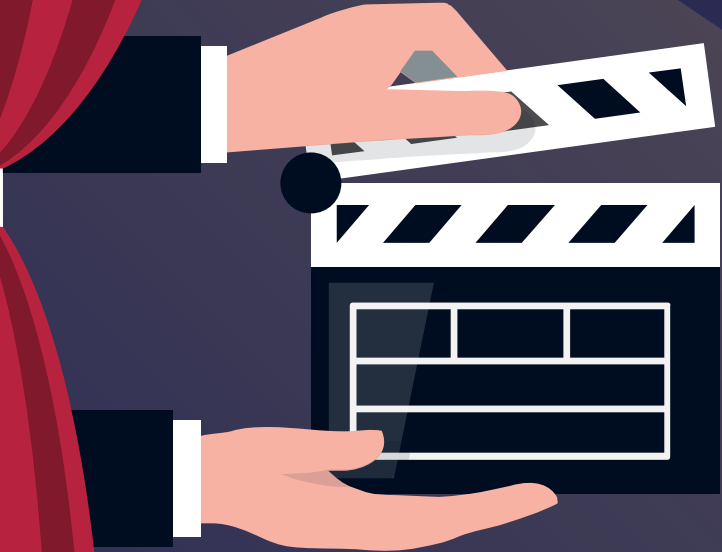
Takeaways

- Binary classification = stable
- Features with strong linear relationships
- Successful before release
- Early green-light decisions
- Supports marketing strategy



Improvements

- Add text-based features
- Try nonlinear models (Random Forest, XGBoost)
- Create more cast/crew quality metrics

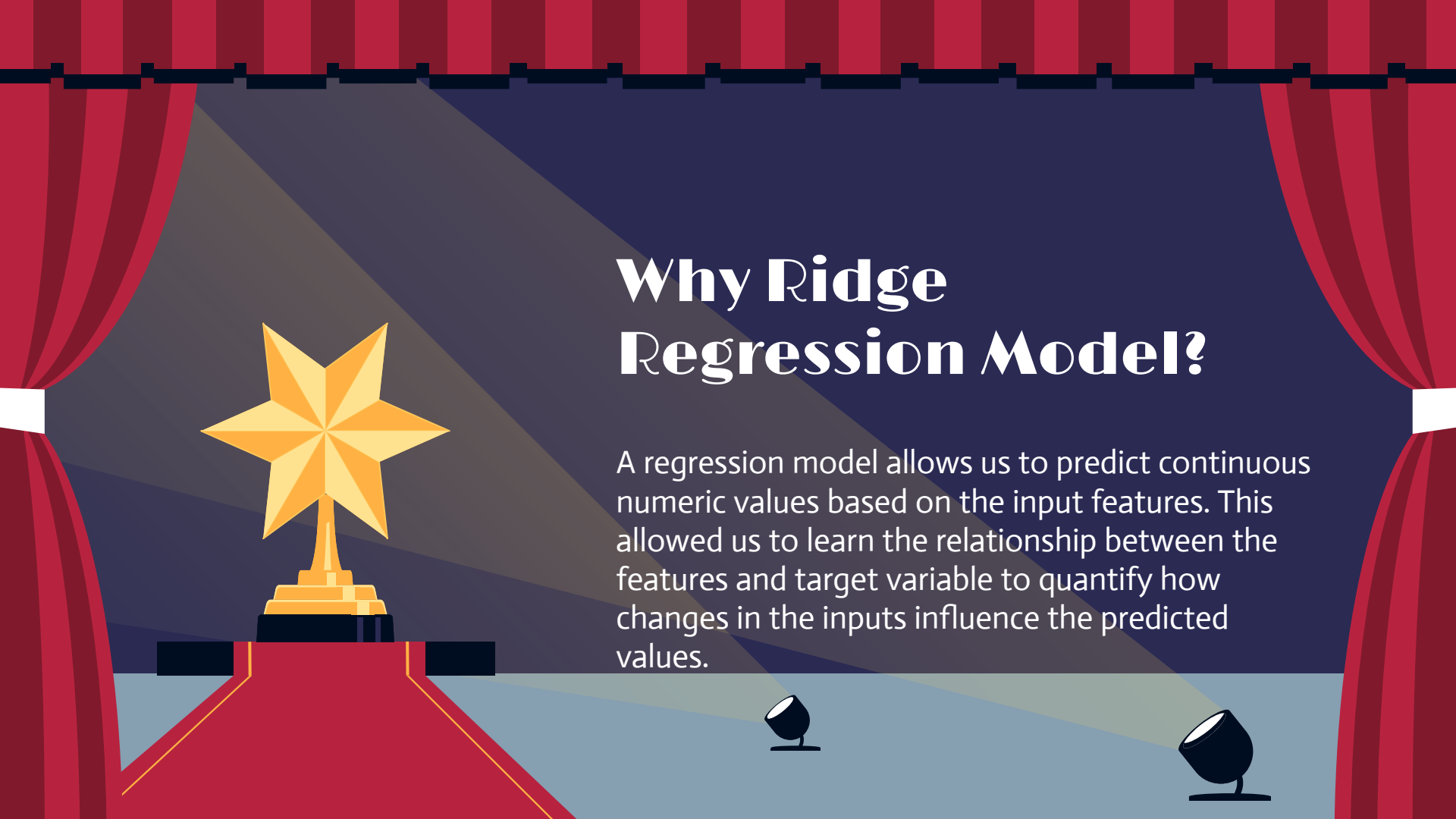
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03

Phase 2

Regression Model,
Takeaways & Improvements

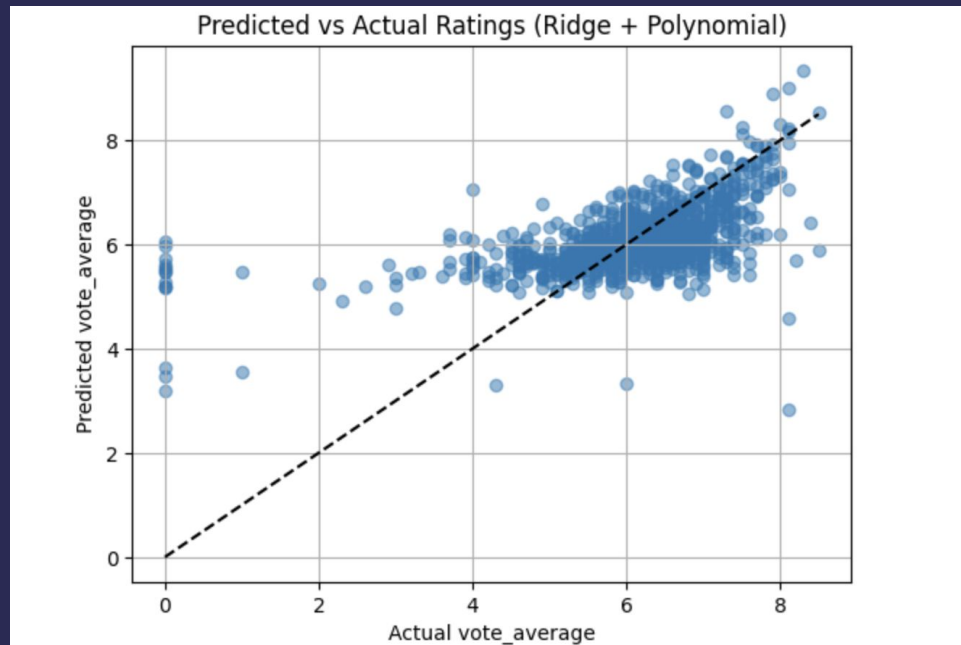
A small, black, stylized stage spotlight on a stand, positioned at the bottom right of the slide.

The background of the slide is a stylized stage. At the top, there are red curtains. A red carpet leads from the bottom left towards a golden trophy. Two spotlights on the floor cast beams of light onto the trophy and the text area. The text is written in white on a dark blue background.

Why Ridge Regression Model?

A regression model allows us to predict continuous numeric values based on the input features. This allowed us to learn the relationship between the features and target variable to quantify how changes in the inputs influence the predicted values.

Regression Model



What does the Ridge Regression Model Tell Us?

Error

The predictions differs by about one rating point, indicating that the model is reasonably accurate.

RSME:
1.0283

Model Tell Us?

Accuracy

Model's prediction closely follows the actual IMDb ratings, with movies falling near the expected value.

MAE:
0.6613

Trend

Higher rated movies tend to be predicted higher as well. This captures the overall positive relationship.

R2:
0.2987

Variability

Certain movies are harder to predict because their ratings are influenced by audience opinions and natural variability in the data.



Ridge Regression cont.



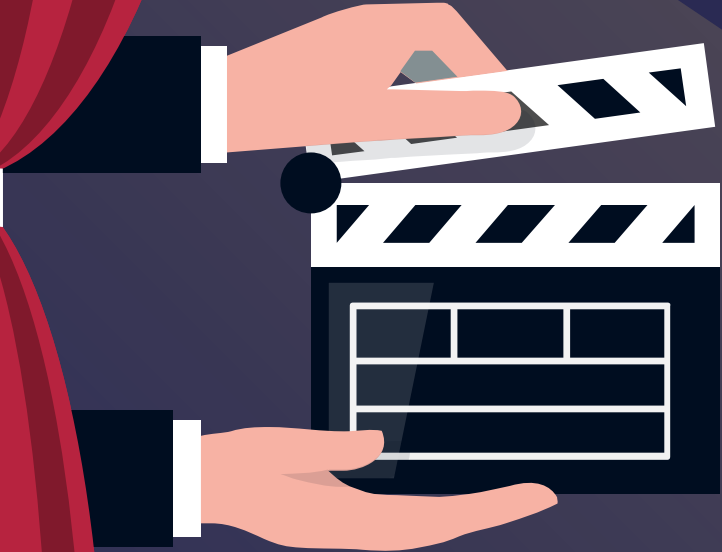
Takeaways

- The model provide a clear estimate of IMDb ratings and follows the true rating trend
- Scatter plot indicates that most predictions fall closely to the reference line.
- Offers a valuable insight into the feasibility of forecasting audience reception for movies.



Improvements

- Refining preprocessing by handling missing values such as filling numeric missing values with the median preventing the extreme values. Eliminate outliers.
- Adjusting Alpha would allow us to get a better understanding the level of regularization to keep the model more stable and product more reliable predictions
- Slight variations of the regression approach to test the different features

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04

Phase 3

Conclusion and Q&A

A small, black, stylized stage spotlight on a stand, pointing upwards towards the center of the slide.

Conclusion

- The Ridge Regression model provides a reliable estimate of the IMDb ratings and reflects the general rating pattern.
- Predictions can help guide marketing decisions such as adjusting campaigns and refining targets when predictions are low.

Thank you!

